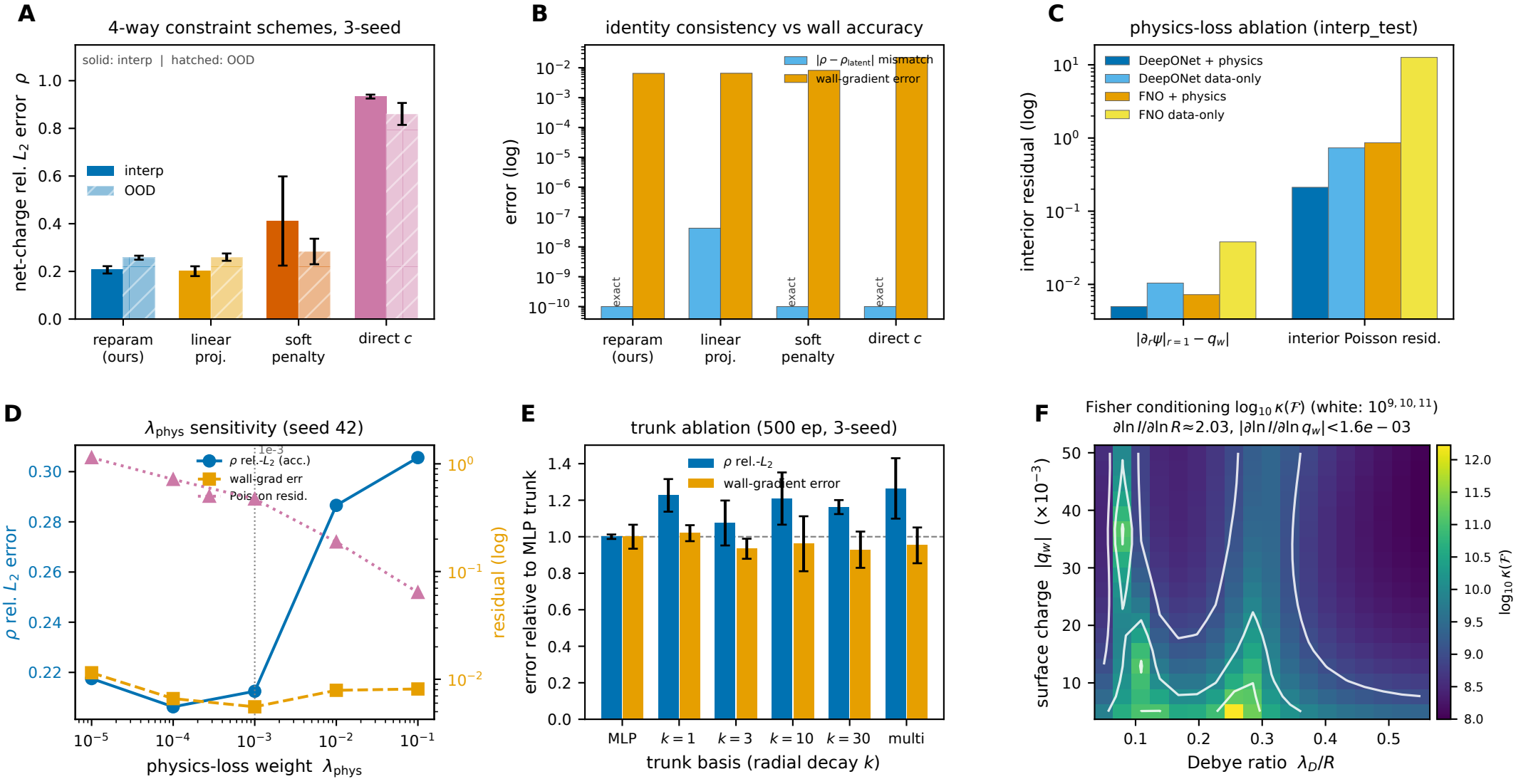
A
4-way constraint schemes, 3-seed
solid: interp | hatched: OOD
net-charge rel. L_2 error ρ
interp
OOD
reparam (ours)
linear proj.
soft penalty
direct c
B
identity consistency vs wall accuracy
error (log)
$|\rho - \rho_{\text{latent}}|$ mismatch
wall-gradient error
exact
reparam (ours)
linear proj.
soft penalty
direct c
C
physics-loss ablation (interp_test)
interior residual (log)
DeepONet + physics
DeepONet data-only
FNO + physics
FNO data-only
$|\partial_r \psi|_{r=1} - q_w|$
interior Poisson resid.
D
λ_{phys} sensitivity (seed 42)
ρ rel.-L_2 (acc.)
wall-grad err
Poisson resid.
1e-3
ρ rel. L_2 error
residual (log)
physics-loss weight λ_{phys}
E
trunk ablation (500 ep, 3-seed)
ρ rel.-L_2
wall-gradient error
error relative to MLP trunk
MLP
$k=1$
$k=3$
$k=10$
$k=30$
multi
trunk basis (radial decay k)
F
Fisher conditioning $\log_{10}\kappa(\mathcal{F})$ (white: $10^{9,10,11}$)
$\partial \ln I/\partial \ln R \approx 2.03$, $|\partial \ln I/\partial \ln q_w| < 1.6e-03$
surface charge $|q_w|$ $(\times 10^{-3})$
Debye ratio λ_D/R
$\log_{10}\kappa(\mathcal{F})$